

OpenCDMi L1 test cases

List of Interfaces of ThunderInterfaces

The Interfaces of thunderinterfaces(IOCDM.h, IDRm.h, IContentDecryption.h) module are inherited and implemented in OpenCDMI Plugin.

The interface details and L1 test cases are added below.

IAccessorOCDM (18 Interfaces) ([ThunderInterfaces/interfaces/IOCDM.h](#) at 1807c13cb4f15265529e4f4e68f5dbd3dc71e06b · [rdkcentral/ThunderInterfaces](#) ([github.com](#)))

In OpenCDMI Plugin, IAccessorOCDM have been implemented and these interface functions are called from Thunderclientlibraries (struct OpenCDMAccessor) module, which is a wrapper to the OpenCDMI.

- CreateSession
- IsTypeSupported
- Metadata
- Metricdata
- CreateSession
- SetServerCertificate
- GetDrmSystemTime
- GetVersionExt
- GetLdlSessionLimit
- IsSecureStopEnabled
- EnableSecureStop
- ResetSecureStops
- GetSecureStopIds
- GetSecureStop
- CommitSecureStop
- DeleteKeyStore
- DeleteSecureStore
- GetKeyStoreHash
- GetSecureStoreHash

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1.	CreateSessionSuccess_withCreateMediaKeySessionSuccess	OCDMImplementation::KeySystem	No	Precondition: OCDMImplementation::Initialize should fill _systemToFactory map. OCDMImplementation::KeySystem will use keySystem /designator and filter the CDMi::IMediaKeys object from _systemToFactory.	::OCDM:: OCDM_RESULT:: OCDM_SUCCESS	CreateSession	const std::string& keySystem, const int32_t licenseType, const std::string& initDataType, const uint8_t* initData, const uint16_t initDataLength, const uint8_t* CDMDData, const uint16_t CDMDDataLength, ::OCDM::ISession::ICallback* callback, std::string& sessionId, ::OCDM::ISession*& session
		CDMi::IMediaKeySession	No	CDMi::IMediaKeySession Pointer named as sessionInterface declared and assigned nullptr.			
		CommonEncryptionData	No	CommonEncryptionData Object created successfully with initData, initDataLength			
		CDMi::IMediaKeys::CreateMediaKeySession	Yes	CreateMediaKeySession will fill the CDMi::IMediaKeySession pointer(sessionInterface) .Implemented in DRM modules (OCDM-Playready/MediaSystem.cpp)			
		Core::Service::Create (SessionImplementation)	No	Creates SessionImplementation object with below arguments. this, keySystem, sessionInterface, callback, &keyIds			
		callback->OnKeyStatusUpdate	Yes	callback is the argument passed by caller function of CreateSession() callbackOnKeyStatusUpdate will sets DRM session status as ::OCDM::ISession::StatusPending			
2	CreateSessionFailed_onCreateMediaKeySessionFailed	CreateMediaKeySession - Implemented in DRM modules (OCDM-Playready/MediaSystem.cpp)	Yes	CreateMediaKeySession will return OCDM::OCDM_RESULT::OCDM_S_FALSE. CreateMediaKeySession fails	OCDM:: OCDM_RESULT:: OCDM_S_FALSE.		
3	CreateSessionFailed_onEmptyKeySystem	OCDMImplementation::KeySystem	No	OCDMImplementation::KeySystem fails.	OCDM:: OCDM_RESULT:: OCDM_S_FALSE.		

4	CreateSessionFailed_onSessionImplementationCreateFailed	Service::Create (SessionImplementation)	No	Service::Create(SessionImplementation) returns fails. Service::Create(SessionImplementation) returns nullptr.	OCDM::OCDM_RESULT::OCDM_S_FALSE.		
5	CreateSessionFailed_onSessionInterfaceIsNull	CreateMediaKeySession	Yes	CreateMediaKeySession will return OCDM::OCDM_RESULT::OCDM_SUCCESS but sessionInterface is a nullptr. CreateMediaKeySession updated sessionInterface as nullptr	OCDM::OCDM_RESULT::OCDM_S_FALSE.		
6	CreateSessionFailed_onEmptyKeyIds	CommonEncryptionData::IsEmpty	No	CommonEncryptionData::IsEmpty returns true.	OCDM::OCDM_RESULT::OCDM_S_FALSE.		

SetServerCertificate

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1	SetServerCertificateSucess_withValidKeySystem	OCDMImplementation::KeySystem	No	Precondition: OCDMImplementation::Initialize should fill _systemToFactory map. OCDMImplementation::KeySystem will use keySystem/designator and filter the CDMi::IMediaKeys object from _systemToFactory.	OCDMImplementation::KeySystem returns CDMi::IMediaKeys object.	SetServerCertificate	const std::string& keySystem, const uint8_t* serverCertificate, const uint16_t serverCertificateLength
		SetServerCertificate	Yes	Implemented in DRM modules (OCDM-Playready/MediaSystem.cpp) CDMi::IMediaKeys	SetServerCertificate will return OCDM::OCDM_RESULT::OCDM_SUCCESS.		
2	setServerCertificateFailed_withInvalidCertificate	SetServerCertificate	Yes	CDMi::IMediaKeys::SetServerCertificate return failure	SetServerCertificate will return OCDM::OCDM_RESULT::OCDM_FAILURE		
3	SetServerCertificateFailed_whenInvalidKeySystem	SetServerCertificate	No	OCDMImplementation::KeySystem fails.	OCDMImplementation::KeySystem returns nullptr.	SetServerCertificate	

GetDrmSystemTime (Need clarification in return value)

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1	GetDrmSystemTimeSucess_withValidKeySystem	OCDMImplementation::KeySystem	No	Precondition: OCDMImplementation::Initialize should fill _systemToFactory map. OCDMImplementation::KeySystem will use keySystem/designator and filter the CDMi::IMediaKeys object from _systemToFactory.	OCDMImplementation::KeySystem returns CDMi::IMediaKeys object.	GetDrmSystemTime	const std::string& keySystem
		GetDrmSystemTime	Yes	Implemented in DRM modules (OCDM-Playready/MediaSystem.cpp) and	GetDrmSystemTime returns OCDM::OCDM_RESULT::OCDM_SUCCESS.		
2	GetDrmSystemFailed_whenInvalidDrmKeySystem	OCDMImplementation::KeySystem	No	OCDMImplementation::KeySystem fails.	OCDMImplementation::KeySystem returns nullptr.	GetDrmSystemTime	
	GetDrmSystemTimeFailed_whenMediaKeysExtGetDrmSystemTimeFailed	GetDrmSystemTime	Yes		GetDrmSystemTime returns OCDM::OCDM_RESULT::OCDM_FAILURE.	GetDrmSystemTime	

GetVersionExt

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1	GetVersionExtSucess	OCDMImplementation::KeySystem Will returns the CDMi::IMediaKeys*	No	Precondition: OCDMImplementation::Initialize should fill _systemToFactory map. OCDMImplementation::KeySystem will use keySystem/designator and filter the CDMi::IMediaKeys object from _systemToFactory.	OCDMImplementation::KeySystem returns CDMi::IMediaKeys object.	GetVersionExt	const std::string& keySystem
		GetVersionExt	Yes	Implemented in DRM modules (OCDM-Playready/MediaSystem.cpp) and returns version	GetVersionExt should return N onempty version value.		
2	GetVersionExtVersionExtEmpty	GetVersionExt	Yes	GetVersionExt returns empty value.	GetVersionExt should return empty version value.	GetVersionExt	

GetLdlSessionLimit

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1	GetLdlSessionLimitSuccess	OCDMImplementation::KeySystem Will returns the CDMi::IMediaKeys*	No	Precondition: OCDMImplementation::Initialize should fill _systemToFactory map. OCDMImplementation::KeySystem will use keySystem /designator and filter the CDMi::IMediaKeys object from _systemToFactory.	OCDMImplementation::KeySystem returns CDMi::IMediaKeys object.	GetLdlSessionLimit	const std::string& keySystem
		GetLdlSessionLimit	Yes	Implemented in DRM modules (<i>OCDM-Playready /MediaSystem.cpp</i>) and returns LdlSessionLimit value.	GetLdlSessionLimit returns valid LdlSessionLimit.		
2	GetLdlSessionLimitLdlSessionLimitEmpty	GetLdlSessionLimit	Yes	GetLdlSessionLimit returns empty.	LdlSessionLimit is zero.	GetLdlSessionLimit	

IsSecureStopEnabled

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1	SecureStopEnabled	OCDMImplementation::KeySystem Will returns the CDMi::IMediaKeys*	No	Precondition: OCDMImplementation::Initialize should fill _systemToFactory map. OCDMImplementation::KeySystem will use keySystem /designator and filter the CDMi::IMediaKeys object from _systemToFactory.	OCDMImplementation::KeySystem returns CDMi::IMediaKeys object.	IsSecureStopEnabled	const std::string& keySystem
		IsSecureStopEnabled	Yes	Implemented in DRM modules (<i>OCDM-Playready /MediaSystem.cpp</i>) and returns true.	IsSecureStopEnabled returns true		
2	SecureStopDisabled	IsSecureStopEnabled	Yes	IsSecureStopEnabled returns false.	IsSecureStopEnabled returns false	IsSecureStopEnabled	

EnableSecureStop

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1	EnableSecureStopEnabled	OCDMImplementation::KeySystem Will returns the CDMi::IMediaKeys*	No	Precondition: OCDMImplementation::Initialize should fill _systemToFactory map. OCDMImplementation::KeySystem will use keySystem /designator and filter the CDMi::IMediaKeys object from _systemToFactory.	OCDMImplementation::KeySystem returns CDMi::IMediaKeys object.	EnableSecureStop	const std::string& keySystem, bool enable
		EnableSecureStop (enable)	Yes	Implemented in DRM modules (<i>OCDM-Playready /MediaSystem.cpp</i>) and returns OCDM::OCDM_RESULT::OCDM_SUCCESS.	OCDM::OCDM_RESULT::OCDM_SUCCESS		
2	EnableSecureStopNotEnabled	EnableSecureStop (enable)	Yes	EnableSecureStop fails.	::OCDM::OCDM_RESULT::OCDM_S_FALSE	EnableSecureStop	

ResetSecureStops

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1	ResetSecureStopsSuccess	OCDMImplementation::KeySystem Will returns the CDMi::IMediaKeys*	No	Precondition: OCDMImplementation::Initialize should fill _systemToFactory map. OCDMImplementation::KeySystem will use keySystem /designator and filter the CDMi::IMediaKeys object from _systemToFactory.	OCDMImplementation::KeySystem returns CDMi::IMediaKeys object.	ResetSecureStops	const std::string& keySystem
		ResetSecureStops	Yes	Implemented in DRM modules (<i>OCDM-Playready /MediaSystem.cpp</i>)	returns valid integer.		
2	ResetSecureStopsFailed	ResetSecureStops	Yes		returns 0.	ResetSecureStops	

GetSecureStopsIds

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
-------	-----------	---	---------------	---------	-----------------	-----------------------	--------------------

1	GetSecureStopIdsSuccess	OCDMImplementation::KeySystem Will returns the CDMi::IMediaKeys*	No	Precondition: OCDMImplementation::Initialize should fill _systemToFactory map. OCDMImplementation::KeySystem will use keySystem /designator and filter the CDMi::IMediaKeys object from _systemToFactory.	OCDMImplementation::KeySystem returns CDMi::IMediaKeys object.	GetSecureStopIds	const std::string& keySystem, unsigned char Ids[], uint16_t idsLength, uint32_t& count
		GetSecureStopIds	Yes	Implemented in DRM modules (<i>OCDM-Playready /MediaSystem.cpp</i>)	::OCDM::OCDM_RESULT::OCDM_SUCCESS		
2	GetSecureStopIdsFailed	GetSecureStopIds	Yes		::OCDM::OCDM_RESULT::OCDM_S_FALSE	GetSecureStopIds	

GetSecureStop

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1	GetSecureStopSuccess	OCDMImplementation::KeySystem Will returns the CDMi::IMediaKeys*	No	Precondition: OCDMImplementation::Initialize should fill _systemToFactory map. OCDMImplementation::KeySystem will use keySystem /designator and filter the CDMi::IMediaKeys object from _systemToFactory.	OCDMImplementation::KeySystem returns CDMi::IMediaKeys object.	GetSecureStop	const std::string& keySystem, const unsigned char sessionId[], uint32_t sessionIdLength, unsigned char* rawData, uint16_t& rawSize
		GetSecureStop	Yes	Implemented in DRM modules (<i>OCDM-Playready /MediaSystem.cpp</i>)	::OCDM::OCDM_RESULT::OCDM_SUCCESS		
2	GetSecureStopFailed	GetSecureStop	Yes		::OCDM::OCDM_RESULT::OCDM_S_FALSE	GetSecureStop	

CommitSecureStop

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1	CommitSecureStopSuccess	OCDMImplementation::KeySystem Will returns the CDMi::IMediaKeys*	No	Precondition: OCDMImplementation::Initialize should fill _systemToFactory map. OCDMImplementation::KeySystem will use keySystem /designator and filter the CDMi::IMediaKeys object from _systemToFactory.	OCDMImplementation::KeySystem returns CDMi::IMediaKeys object.	CommitSecureStop	const std::string& keySystem, const unsigned char sessionId[], uint32_t sessionIdLength, const unsigned char serverResponse[], uint32_t serverResponseLength
		CommitSecureStop	Yes	Implemented in DRM modules (<i>OCDM-Playready /MediaSystem.cpp</i>)	::OCDM::OCDM_RESULT::OCDM_SUCCESS		
2	CommitSecureStopFailed	CommitSecureStop	Yes		::OCDM::OCDM_RESULT::OCDM_S_FALSE	CommitSecureStop	

DeleteKeyStore

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1	DeleteKeyStoreSuccess	OCDMImplementation::KeySystem Will returns the CDMi::IMediaKeys*	No	Precondition: OCDMImplementation::Initialize should fill _systemToFactory map. OCDMImplementation::KeySystem will use keySystem /designator and filter the CDMi::IMediaKeys object from _systemToFactory.	OCDMImplementation::KeySystem returns CDMi::IMediaKeys object.	DeleteKeyStore	const std::string& keySystem
		DeleteKeyStore	Yes	Implemented in DRM modules (<i>OCDM-Playready /MediaSystem.cpp</i>)	::OCDM::OCDM_RESULT::OCDM_SUCCESS		
2	DeleteKeyStoreFailed	DeleteKeyStore	Yes		::OCDM::OCDM_RESULT::OCDM_S_FALSE	DeleteKeyStore	

DeleteSecureStore

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
-------	-----------	---	---------------	---------	-----------------	-----------------------	--------------------

1	DeleteSecureStoreSuccess	OCDMImplementation::KeySystem Will returns the CDMi::IMediaKeys*	No	Precondition: OCDMImplementation::Initialize should fill _systemToFactory map. OCDMImplementation::KeySystem will use keySystem /designator and filter the CDMi::IMediaKeys object from _systemToFactory.	OCDMImplementation::KeySystem returns CDMi::IMediaKeys object.	DeleteSecureStore	const std::string&keySystem
		DeleteSecureStore	Yes	Implemented in DRM modules (<i>OCDM-Playready /MediaSystem.cpp</i>)	::OCDM::OCDM_RESULT::OCDM_SUCCESS		
2	DeleteSecureStoreFailed	DeleteSecureStore	Yes		::OCDM::OCDM_RESULT::OCDM_S_FALSE	DeleteSecureStore	

GetKeyStoreHash

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1	GetSecureStoreHashSuccess	OCDMImplementation::KeySystem Will returns the CDMi::IMediaKeys*	No	Precondition: OCDMImplementation::Initialize should fill _systemToFactory map. OCDMImplementation::KeySystem will use keySystem /designator and filter the CDMi::IMediaKeys object from _systemToFactory.	OCDMImplementation::KeySystem returns CDMi::IMediaKeys object.	GetSecureStoreHash	const std::string&keySystem, uint8_t keyStoreHash[], uint32_t keyStoreHashLength
		GetSecureStoreHash	Yes	Implemented in DRM modules (<i>OCDM-Playready /MediaSystem.cpp</i>)	::OCDM::OCDM_RESULT::OCDM_SUCCESS		
2	GetSecureStoreHashFailed	GetSecureStoreHash	Yes		::OCDM::OCDM_RESULT::OCDM_S_FALSE	GetSecureStoreHash	

GetSecureStoreHash

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1	GetSecureStoreHashSuccess	OCDMImplementation::KeySystem Will returns the CDMi::IMediaKeys*	No	Precondition: OCDMImplementation::Initialize should fill _systemToFactory map. OCDMImplementation::KeySystem will use keySystem /designator and filter the CDMi::IMediaKeys object from _systemToFactory.	OCDMImplementation::KeySystem returns CDMi::IMediaKeys object.	GetSecureStoreHash	const std::string&keySystem, uint8_t secureStoreHash[], uint32_t secureStoreHashLength
		GetSecureStoreHash	Yes	Implemented in DRM modules (<i>OCDM-Playready /MediaSystem.cpp</i>)	::OCDM::OCDM_RESULT::OCDM_SUCCESS		
2	GetSecureStoreHashFailed	GetSecureStoreHash	Yes		::OCDM::OCDM_RESULT::OCDM_S_FALSE	GetSecureStoreHash	

IMediaKeySessionCallback(4 Interfaces) (ThunderInterfaces/interfaces/IDRM.h at 1807c13cb4f15265529e4f4e68f5dbd3dc71e06b · rdkcentral /ThunderInterfaces (github.com))

- OnKeyMessage
- OnError
- OnKeyStatusUpdate
- OnKeyStatusesUpdated

OnKeyMessage

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1	OnKeyMessage Success	_callback->OnKeyMessage	Yes	PreCondition: Should call Sink::Sink to fill _callback value.	Returns void value.	OnKeyMessage	const uint8_t* f_pbKeyMessage, uint32_t f_cbKeyMessage, const char* f_pszUrl
2.	OnKeyMessage Failed		No	_callback is nullptr. Not able to trigger OnKeyMessage function.	Returns void value.	OnKeyMessage	

OnError

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1	OnErrorSuccess	_callback->OnError	Yes	PreCondition: Should call Sink::Sink to fill _callback value.	Returns void value.	OnError	int16_t f_nError,

2.	OnErrorFailed		No	_callback is nullptr. Not able to trigger OnError function.	Returns void value.		CDMI::CDMI_RESULT f_crSysError, const char* errorMessage
----	---------------	--	----	---	---------------------	--	--

OnKeyStatusUpdate

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1.	OnKeyStatusUpdateFailed	CommonEncryptionData::KeyId (CommonEncryptionData::COMMON, buffer, length) SessionImplementation::CommonEncryptionData::UpdateKeyStatus (key) _callback->OnKeyStatusUpdate	No No Yes	PreCondition: Shouldn't call Sink::Sink to fill _callback value. _callback is nullptr. Not able to trigger OnKeyStatusUpdate function.	keystatus = InternalError	OnKeyStatusUpdate	const char* keyMessage, const uint8_t buffer[], const uint8_t length
2.	OnKeyStatusUpdateKeyMessageIsKeyUsable	CommonEncryptionData::KeyId (CommonEncryptionData::COMMON, buffer, length) SessionImplementation::CommonEncryptionData::UpdateKeyStatus (key) _callback->OnKeyStatusUpdate	No No Yes	Updates the key status as MessageIsKeyUsable PreCondition: Should call Sink::Sink to fill _callback value.	keystatus = MessageIsKeyUsable		
3.	OnKeyStatusUpdateKeyReleased	CommonEncryptionData::KeyId (CommonEncryptionData::COMMON, buffer, length) SessionImplementation::CommonEncryptionData::UpdateKeyStatus (key) _callback->OnKeyStatusUpdate	No No Yes	PreCondition: Should call Sink::Sink to fill _callback value.	keystatus = KeyReleased		
4.	OnKeyStatusUpdateKeyExpired	CommonEncryptionData::KeyId (CommonEncryptionData::COMMON, buffer, length) SessionImplementation::CommonEncryptionData::UpdateKeyStatus (key) _callback->OnKeyStatusUpdate	No No Yes	PreCondition: Should call Sink::Sink to fill _callback value.	keystatus = KeyExpired		
5.	OnKeyStatusUpdateKeyOutputRestricted	CommonEncryptionData::KeyId (CommonEncryptionData::COMMON, buffer, length) SessionImplementation::CommonEncryptionData::UpdateKeyStatus (key) _callback->OnKeyStatusUpdate	No No Yes	PreCondition: Should call Sink::Sink to fill _callback value.	keystatus = KeyOutputRestricted .		
6.	OnKeyStatusUpdateKeyOutputDownscaled	CommonEncryptionData::KeyId (CommonEncryptionData::COMMON, buffer, length) SessionImplementation::CommonEncryptionData::UpdateKeyStatus (key) _callback->OnKeyStatusUpdate	No No Yes	PreCondition: Should call Sink::Sink to fill _callback value.	keystatus = KeyOutputDownscaled		
7.	OnKeyStatusUpdateSEC_RESULT_HW_FAILURE	CommonEncryptionData::KeyId (CommonEncryptionData::COMMON, buffer, length) SessionImplementation::CommonEncryptionData::UpdateKeyStatus (key) _callback->OnKeyStatusUpdate	No No Yes	PreCondition: Should call Sink::Sink to fill _callback value.	keystatus = SEC_RESULT_HW_FAILURE .		
8.	OnKeyStatusUpdateKeyOutputRestrictedHDCP22	CommonEncryptionData::KeyId (CommonEncryptionData::COMMON, buffer, length) SessionImplementation::CommonEncryptionData::UpdateKeyStatus (key) _callback->OnKeyStatusUpdate	No No Yes	PreCondition: Should call Sink::Sink to fill _callback value.	keystatus = KeyOutputRestrictedHDCP22		
9.	OnKeyStatusUpdateInvalidKeyMessage	CommonEncryptionData::KeyId (CommonEncryptionData::COMMON, buffer, length) SessionImplementation::CommonEncryptionData::UpdateKeyStatus (key) _callback->OnKeyStatusUpdate	No No Yes	PreCondition: Should call Sink::Sink to fill _callback value.	keystatus = InvalidKeyMessage		

OnKeyStatusesUpdated

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1	OnKeyStatusesUpdatedSuccess	_callback->OnKeyStatusesUpdated	Yes	PreCondition: Should call Sink::Sink to fill _callback value.	Returns void value.	OnKeyStatusesUpdated	No arguments
2.	OnKeyStatusesUpdatedFailed		No	_callback is nullptr. Not able to trigger OnKeyStatusesUpdated function.	Returns void value.		

ISessionExt (8 Interfaces) ([ThunderInterfaces/interfaces/IOCDM.h at 1807c13cb4f15265529e4f4e68f5dbd3dc71e06b · rdkcentral/ThunderInterfaces \(github.com\)](#))

- SessionIdExt
- BufferIdExt
- SetDrmHeader
- GetChallengeDataExt
- CancelChallengeDataExt
- StoreLicenseData
- SelectKeyId
- CleanDecryptContext

SessionIdExt

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1.	SessionIdExtReturnsValidSessionId	GetSessionIdExt()	Yes	implemented in OCDM-Playready /MediaSessionExt.cpp	returns valid SessionId value.	SessionIdExt	No Arguments
2.	SessionIdExtReturnsZero	GetSessionIdExt()	Yes	implemented in OCDM-Playready /MediaSessionExt.cpp	returns zero value.		

BufferIdExt (Covered SessionImplementation::BufferId() interface Api)

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1.	BufferIdExt Success	BufferId()	No	BufferId() returns _bufferName(), which is a string. _buffer is a DataExchange object. _buffer is initialized in SessionImplementation::CreateSessionBuffer()	BufferId() will return valid string.	BufferIdExt	No Arguments
2.	BufferIdExt Failed	BufferId()	No	BufferId() returns empty string as _buffer is nullptr.	BufferId() returns empty string.		

SetDrmHeader

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1.	SetDrmHeader Success	SetDrmHeader()	Yes	implemented in OCDM-Playready /MediaSessionExt.cpp	SetDrmHeader returns CDMi_SUCCESS.	SetDrmHeader	const uint8_t drmHeader[],
2.	SetDrmHeader Failed	SetDrmHeader()	Yes	implemented in OCDM-Playready /MediaSessionExt.cpp	SetDrmHeader returns CDMi_S_FALSE.		uint32_t drmHeaderLength

GetChallengeDataExt

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1.	GetChallengeDataExtSuccess	GetChallengeDataExt()	Yes	implemented in OCDM-Playready/MediaSessionExt.cpp	GetChallengeDataExt returns CDMi_SUCCESS.	GetChallengeDataExt	uint8_t* challenge,
2.	GetChallengeDataExtFailed	GetChallengeDataExt()	Yes	implemented in OCDM-Playready/MediaSessionExt.cpp	GetChallengeDataExt returns CDMi_S_FALSE.		uint32_t& challengeSize,
3.	GetChallengeDataExtFailedWithoutMemory	GetChallengeDataExt()	Yes	implemented in OCDM-Playready/MediaSessionExt.cpp	GetChallengeDataExt returns CDMi_OUT_OF_MEMORY		uint32_t isLDL

CancelChallengeDataExt

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1.	CancelChallengeDataExtSuccess	CancelChallengeDataExt()	Yes	implemented in OCDM-Playready /MediaSessionExt.cpp	CancelChallengeDataExt returns CDMi_SUCCESS.	CancelChallengeDataExt	No Arguments
2.	CancelChallengeDataExtFailed	CancelChallengeDataExt()	Yes	implemented in OCDM-Playready /MediaSessionExt.cpp	CancelChallengeDataExt returns CDMi_S_FALSE.		

StoreLicenseData

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1.	StoreLicenseDataSuccess	StoreLicenseData()	Yes	implemented in OCDM-Playready /MediaSessionExt.cpp	StoreLicenseData returns CDMi_SUCCESS.	StoreLicenseData	const uint8_t licenseData[],
2.	StoreLicenseDataFailed	StoreLicenseData()	Yes	implemented in OCDM-Playready /MediaSessionExt.cpp	StoreLicenseData returns CDMi_S_FALSE.		uint32_t licenseDataSize, unsigned char* secureStopId

SelectKeyId

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1.	SelectKeyIdSuccess	SelectKeyId()	Yes	implemented in OCDM-Playready /MediaSessionExt.cpp	SelectKeyId returns CDMi_SUCCESS.	SelectKeyId	const uint8_t keyLength,
2.	SelectKeyIdFailed	SelectKeyId()	Yes	implemented in OCDM-Playready /MediaSessionExt.cpp	SelectKeyId returns CDMi_S_FALSE.		const uint8_t keyId[]

CleanDecryptContext

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1.	CleanDecryptContextSuccess	CleanDecryptContext()	Yes	implemented in OCDM-Playready /MediaSessionExt.cpp	CleanDecryptContext returns CDMi_SUCCESS.	CleanDecryptContext	No Arguments
2.	CleanDecryptContextFailed	CleanDecryptContext()	Yes	implemented in OCDM-Playready /MediaSessionExt.cpp	CleanDecryptContext returns CDMi_S_FALSE.		

ISession (13 Interfaces) ([ThunderInterfaces/interfaces/IOCDM.h at 1807c13cb4f15265529e4f4e68f5dbd3dc71e06b · rdkcentral/ThunderInterfaces \(github.com\)](#))

- Load
- Update
- Remove
- Metadata
- Metricdata
- Status
- Status
- CreateSessionBuffer
- BufferId
- SessionId
- Close
- ResetOutputProtection
- Revoke

Load

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1.	LoadSuccess	MediaKeySession::Load()	Yes	implemented in OCDM-Playready /MediaSession.cpp	Load returns CDMi_SUCCESS.	Load	No Arguments
2.	LoadFailed	MediaKeySession::Load()	Yes	implemented in OCDM-Playready /MediaSession.cpp	Load returns CDMi_S_FALSE.		

Update

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1.	UpdateSuccess	Update()	Yes	implemented in OCDM-Playready /MediaSession.cpp	returns void.	Update	const uint8_t* keyMessage, const uint16_t keyLength

Remove

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1.	RemoveSuccess	Remove()	Yes	implemented in OCDM-Playready /MediaSession.cpp	Remove returns CDMI_SUCCESS.	Remove	No Arguments
2.	RemoveFailed	Remove()	Yes	implemented in OCDM-Playready /MediaSession.cpp	Remove returns CDMI_S_FALSE.		

Metadata

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1.	MetadataSuccess	GetMetadata()	Yes	implemented in OCDM-Playready /MediaSession.cpp	GetMetadata returns valid string.	Metadata	No Arguments
2.	MetadataFailed	GetMetadata()	Yes	implemented in OCDM-Playready /MediaSession.cpp	GetMetadata returns empty Metadata.		

Metricdata (Not implemented here)

Status without arguments

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1.	StatusSuccess	CommonEncryptionData::Status()	No	Need to update CommonEncryptionData::_keyIds list before starting test.	Status returns ISession::KeyStatus::Usable	Status	No Arguments
2.	StatusFailedWithStatusPending	CommonEncryptionData::Status()	No	Need to update CommonEncryptionData::_keyIds list before starting test.	Status returns ISession::KeyStatus::StatusPending		

Status with arguments

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1.	StatusSuccess	CommonEncryptionData::Status (with arguments)	No	Need to update CommonEncryptionData::_keyIds list before starting test.	CommonEncryptionData::Status() returns ISession::KeyStatus::Usable	Status	const uint8_t keyId[], const uint8_t length
		OCDM::KeyId::IsValid()	No		IsValid() returns True.		
2.	StatusInvalidKey	OCDM::KeyId::IsValid()	No	Need to update CommonEncryptionData::_keyIds list before starting test.	IsValid() returns False. So, CommonEncryptionData::Status() returns ISession::KeyStatus::StatusPending		
3.	StatusInputKeyIsNotInList	CommonEncryptionData::Status (with arguments)	No	Need to update CommonEncryptionData::_keyIds list before starting test.	Given input key is not in the CommonEncryptionData::_keyIds list.		

CreateSessionBuffer

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1.	CreateSessionBufferSuccess	Core::CriticalSection::Lock(), Core::CriticalSection::Unlock()	No			CreateSessionBuffer	std::string& bufferID
		AccessorOCDM::BufferAdministrator::AcquireBuffer()	No		Returns true.		
		DataExchange()	No		Creates DataExchange object with given bufferID		
					CreateSessionBuffer returns ::OCDM::OCDM_SUCCESS		
2.	CreateSessionBufferNotNull		No		CreateSessionBuffer() fills bufferID and returns ::OCDM::OCDM_S_FALSE		
3.	CreateSessionBufferInvalidBufferID	AccessorOCDM::BufferAdministrator::AcquireBuffer()	No		Returns false. CreateSessionBuffer returns ::OCDM::OCDM_INVALID_DECRYPT_BUFFER		

BufferId - Test cases covered as part of ISessionExt::BufferIdExt()

SessionId

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1.	SessionIdSucess			SessionImplementation object should be created with proper _sessionId value.	returns valid _sessionId string.	CleanDecryptContext	No Arguments
2.	SessionIdReturnEmptyValue				returns empty string.		

Close

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1.	CloseSucess	Close()	Yes	implemented in OCDM-Playready /MediaSession.cpp	Close() returns void	Close	No Arguments

ResetOutputProtection

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1.	ResetOutputProtectionSucess	ResetOutputProtection()	Yes	implemented in OCDM-Playready /MediaSession.cpp	ResetOutputProtection returns CDMi_SUCCESS.	ResetOutputProtection	No Arguments
2.	ResetOutputProtectionFailed	ResetOutputProtection()	Yes	implemented in OCDM-Playready /MediaSession.cpp	ResetOutputProtection returns CDMi_S_FALSE.		

Revoke(Internally called Sink::Revoke() api)

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1.	RevokeSucess	Sink::Revoke()	No	callback should not be nullptr.	_callback need to have valid data	Revoke()	OCDM::ISession::ICallback* callback
		ISession::ICallback::Release()	No				
2.	RevokeCallbackNullptr	CleanDecryptContext()	Yes	callback should be nullptr.	_callback have valid data.		

IContentDecryption (6 Interfaces) (ThunderInterfaces/interfaces/IContentDecryption.h at 1807c13cb4f15265529e4f4e68f5dbd3dc71e06b · rdkcentral/ThunderInterfaces (github.com))

- Initialize
- Deinitialize
- Reset
- Systems
- Designators
- Sessions

Initialize

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1	InitializeSucess	PluginHost::IShell::SubSystems	No	gets SubSystems	Returns valid subSystem	Initialize	PluginHost::IShell* service
		PluginHost::IShell::ConfigLine	No	loading the configured factories			
		PluginHost::IShell::DataPath	No	gets DataPath.			
		WPEFramework::Plugin::Config	No	loads opencdm configuration			
		Core::Directory	No	Core::Directory object created			
		Core::Library	No	Core::Library object created			
		Core::Library::IsLoaded	No				
		Core::Library::LoadFunction	No				
		CDMi::ISystemFactory	No	CDMi::ISystemFactory object created			

		SystemFactory	No	SystemFactory object created	
		CDMi::ISystemFactory::KeySystem()	No		
		Core::File::FileNameExtended	No		
		Core::Library::Error()	No		
		Core::JSON::ArrayType	No		
		Core::JSON::String	No		
		::OCDM::IAccessorOCDM	No		
		RPC::InvokeServer	No		
		Core::IWorkerPool::Instance	No		
		PluginHost::IShell::SubSystems::IsActive	No	verifying SubSystems status.	
		PluginHost::IShell::SubSystems::Set	No		
2	InitializeSubSystemsNull	PluginHost::IShell::SubSystems	No	service has null SubSystems.	Returns null pointer
3.	InitializeLoadingConfigFailed	WPEFramework::Plugin::Config::FromString	No	FromString returns empty string	returns empty string
4	InitializeIsLoadedFailed	Core::Library::IsLoaded	No	IsLoaded returns false	returns false
5.	InitializeLoadFunctionFailed	Core::Library::LoadFunction	No	LoadFunction returns nullptr	
6.	InitializeDesignatorsIsSetFalse	Designators.IsSet()	No	returns false	
7.	InitializeDesignatorsEmpty	Designators.Elements()	No	returns empty Designators.	
8.	InitializeSubSystemsIsActiveFalse	PluginHost::IShell::SubSystems::IsActive	No	returns false	returns false

Deinitialize

S. No	Test Case	Functions internally used by testing function	Required Mock	Remarks	Expected Result	Testing Function Name	Function arguments
1	DeinitializeSuccess	CDMi::ISystemFactory::Deinitialize	yes	Precondition: Need to fill _systemToFactory	_systemToFactory has valid data and it should become empty.	Deinitialize	PluginHost::IShell* service
2.	DeinitializeSystemFactoryIsEmpty	CDMi::ISystemFactory::Deinitialize	yes	No need to fill _systemToFactory	_systemToFactory has no entries.		

Json RPC Interface APIs communicating through Thunder.

L1 Use Cases for Json RPC Interface APIs

- get_drms
- get_keysystems

S. No	API Name	Test Case	Curl Request	Curl Response	Expected Result	Function arguments
1	get_drms	DrmSystemsAreNotAvailable	{ "jsonrpc": "2.0", "id": 42, "method": "OCDM.drms" }	{ "jsonrpc": "2.0", "id": 42, "result": [] }	string result = []	Core::JSON::ArrayType<DrmData>& response
		AvailableOnlyPlayReadyDrmSystem	{ "jsonrpc": "2.0", "id": 42, "method": "OCDM.drms" }	{ "jsonrpc": "2.0", "id": 42, "result": [{ "name": "PlayReady", "keysystems": ["com.microsoft.playready"] }] }	string result = [{ "name": "PlayReady", "keysystems": ["com.microsoft.playready"] }] }	
		AvailableOnlyWidevineDrmSystem	TBD	TBD	TBD	
		DrmSystemswithlongname	TBD	TBD	TBD	
2.	get_keysystems	GetKeysystemsOfPlayReady	{ "jsonrpc": "2.0", "id": 42, "method": "OCDM.keysystems@PlayReady" }	{ "jsonrpc": "2.0", "id": 42, "result": ["com.microsoft.playready"] }	string result = ["com.microsoft.playready"]	const string& index, Core::JSON::ArrayType<Core::JSON::String>& response

	GetKeysystemsOfWidevine	TBD	TBD	TBD
	NoKeysystemsForPlayReady	{ "jsonrpc": "2.0", "id": 42, "method": "OCDM.keysystems@PlayReady" }	{ "jsonrpc": "2.0", "id": 42, "result": [] }	string result = []
	NoKeysystemsForWidevine	TBD	TBD	TBD
	InvalidInput	TBD	TBD	TBD

Exploring Json RPC interface APIs implementation

Implemented in OCDMJsonRpc.cpp file.

Testing Functions	Functions internally used by testing function	Required Mock	Remarks
get_drms Adds all the available DRMs and their keySystems to the response.	OCDMImplementation::Systems()	No	Returns the available DRM Systems to calling function. This function converting _keySystems member variable from type string to type Core::Service<RPC::StringIterator>::Create<RPC::StringIterator> and returning to calling function. This api is available in FrameworkRPC.cpp file, available in OpenCdm plugin.
	RPC::IteratorType::Next() code: while (drmsIter->Next(element))	NO	Executing while loop for all available DRM Systems. (Eg: Playready) This code is in Thunder/Source/com/ICOM.h
	KeySystems()	NO	Returns list of available keySystems/designators for given DRM Systems. Available in same file.
	OCDMImplementation::Designators(const string& keySystem) code: RPC::StringIterator* keySystemsIter (_opencdmi->Designators(name));	NO	Wrapper function to get keySystems/designators for given DRM Systems. This api is available in FrameworkRPC.cpp file, available in OpenCdm plugin.
	OCDMImplementation::LoadDesignators (const string& keySystem, std::list<string>& designators)	NO	Returns list of available keySystems/designators for given DRM Systems. This api is available in FrameworkRPC.cpp file, available in OpenCdm plugin.
get_keySystems Adds all the available KeySystems for given DRM System to the response.	KeySystems()	NO	Returns list of available keySystems/designators for given DRM Systems. Available in same file.
	OCDMImplementation::Designators(const string& keySystem) code: RPC::StringIterator* keySystemsIter (_opencdmi->Designators(name));	NO	Wrapper function to get keySystems/designators for given DRM Systems. This api is available in FrameworkRPC.cpp file, available in OpenCdm plugin.
	OCDMImplementation::LoadDesignators (const string& keySystem, std::list<string>& designators)	NO	Returns list of available keySystems/designators for given DRM Systems. This api is available in FrameworkRPC.cpp file, available in OpenCdm plugin.

List of mocks to be created for the testing the OpenCDMi Plugin APIs

[OpenCDMi_MockCreationData.xlsx](#)

Challenges Faced in L1 Unit Testing for OpenCDMi:

Initial Test Case Creation:

Started by creating test cases for all OpenCDMi plugin interfaces.

Prototype Limitation:

Faced a limitation where the prototypes of these interfaces were unexposed, preventing direct calls from the test framework.

RPC Implementation in Thunder:

Investigated how unexposed OpenCDMi interfaces are used in the real system.

Discovered that Thunder client libraries utilize Remote Procedure Call (RPC) to access OpenCDMi interfaces.

The Thunder client libraries use an interface pointer, `_remote`, instead of directly creating an instance of `AccessorOCDM`.

This pointer references an object that implements the `IAccessorOCDM` interface.

When the client invokes `CreateSession`, it calls the method accordingly.

This way Thunder client able to trigger `CreateSession` on a remote object without direct access to `AccessorOCDM`.

https://github.com/rdkcentral/ThunderClientLibraries/blob/master/Source/ocdm/open_cdm_impl.h#L87

Example:

```
mutable Core::ProxyType<RPC::CommunicatorClient> _client;  
mutable Exchange::IAccessorOCDM* _remote;  
_remote = _client->Open<Exchange::IAccessorOCDM>(_T("OpenCDMImplementation"));
```

Verification Approach:

Trying an approach to verify Thunder client libraries' functionality from the L1 test framework.

Trying to access OpenCDMi interfaces indirectly through Thunder client libraries using existing RPC mechanisms.

Planning to indirectly verify the functionality of OpenCDMi interfaces through Thunder client libraries using RPC.